

TTK4240 Industriell elektroteknikk

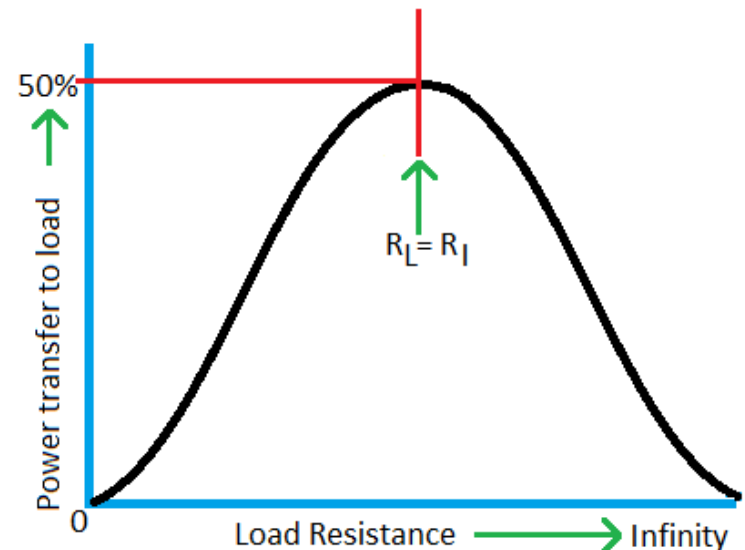
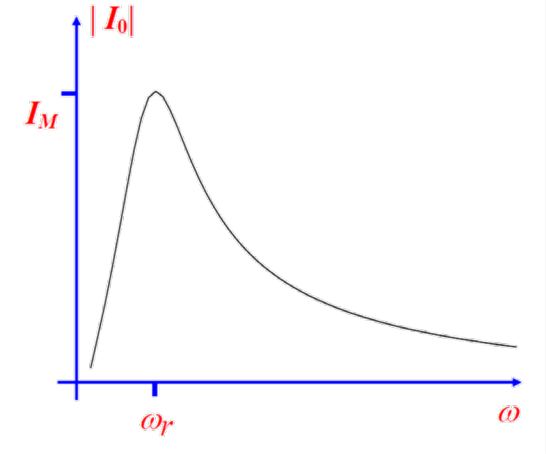
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Lecture - Week 05

- Exercises on power calculations
 - Maximum real power delivered to a load
 - Load impedance required to deliver maximum real power to the load
 - Restrictive loads



Practical examples

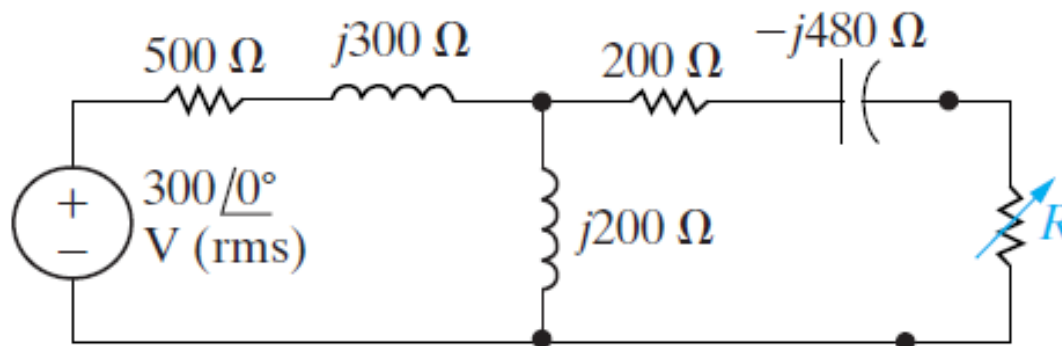
Exercise

Exercise (P.46)

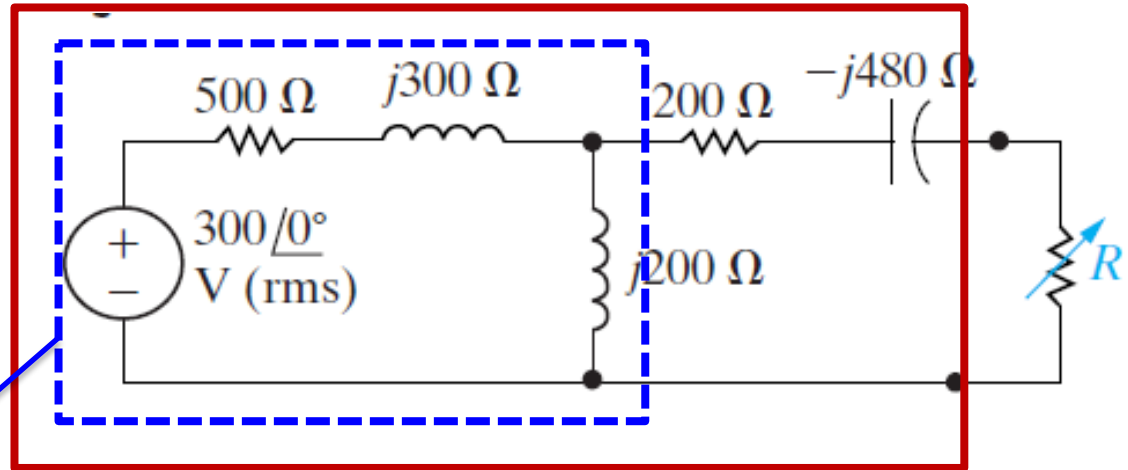
The **variable resistor** in the circuit shown in Fig. P10.46 is adjusted until the average power absorbed is maximum.

- a) Find R .
- b) Find the maximum average power.
- c) Find a resistor in Appendix H that would have the most average power delivered to it.

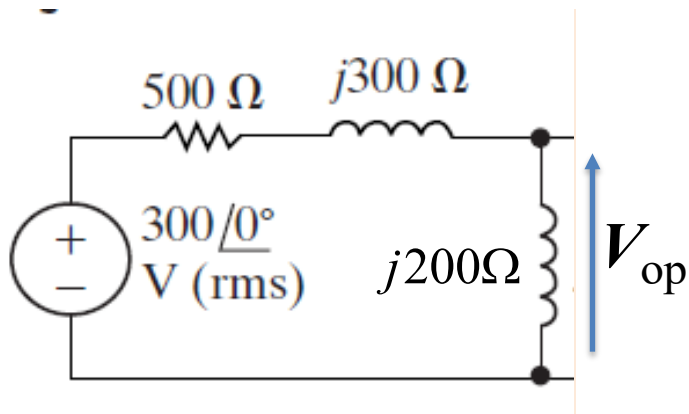
Figure P10.46



Find the Thevenin equivalent?



For simpler analysis, we can find the Thevenin equivalent of the blue box first



$$V_{op} = \frac{300\angle 0^\circ}{500 + 300j + 200j} \times 200j$$

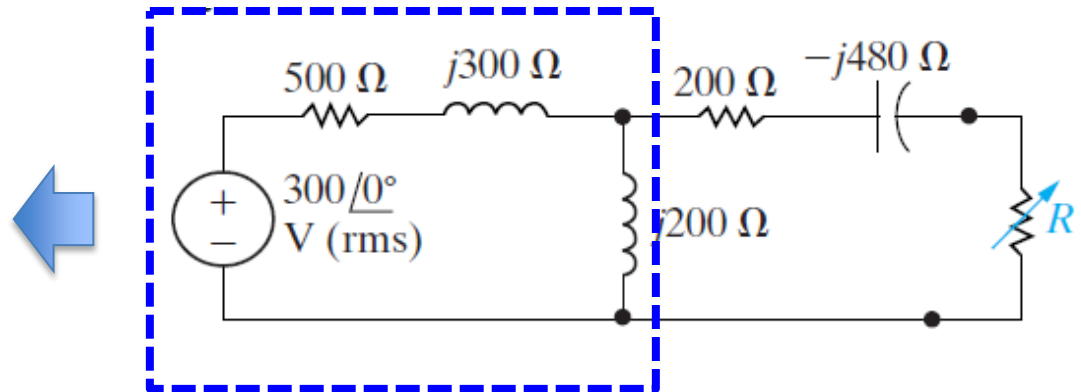
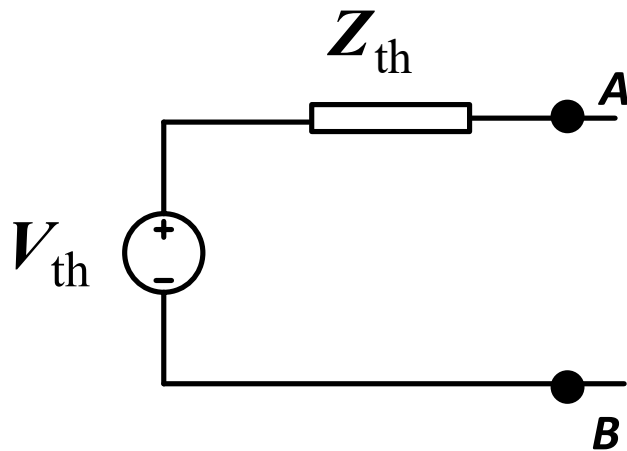
$$= 60\sqrt{2}\angle 45^\circ$$

$$I_{sc} = \frac{300\angle 0^\circ}{500 + 300j} = 0.514\angle -30.96^\circ$$

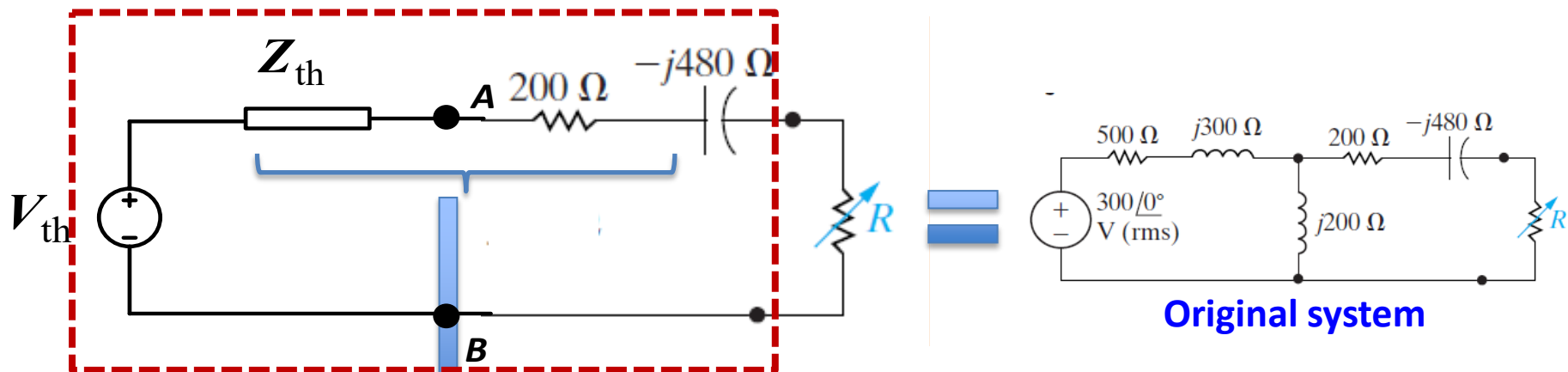
Thevenin equivalent

Thevenin Equivalent
voltage and current

$$\left\{ \begin{array}{l} V_{th} = V_{op} = 60\sqrt{2}\angle 45^\circ \\ Z_{th} = \frac{V_{th}}{I_{sc}} = \frac{60\sqrt{2}\angle 45^\circ}{0.514\angle -30.96^\circ} = 165\angle 75.96^\circ \end{array} \right.$$



Redefine the Thevenin equivalent



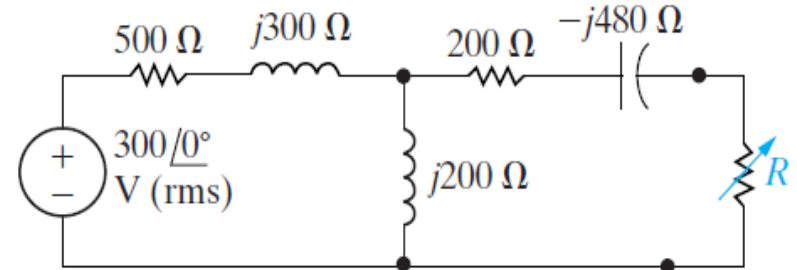
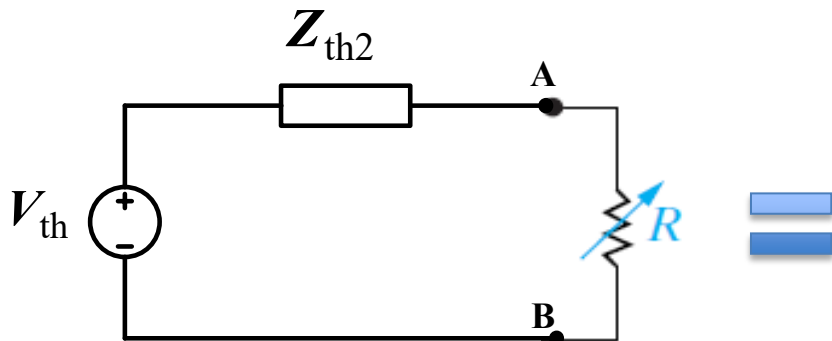
The **NEW** Thevenin Equivalent voltage and impedance

$$V_{th} = V_{op} = 60\sqrt{2}\angle 45^\circ$$

$$Z_{th2} = Z_{th} + 200\Omega - 480j\Omega$$

$$= (240 - 320j)\Omega$$

The **New** Thevenin equivalent



Original system

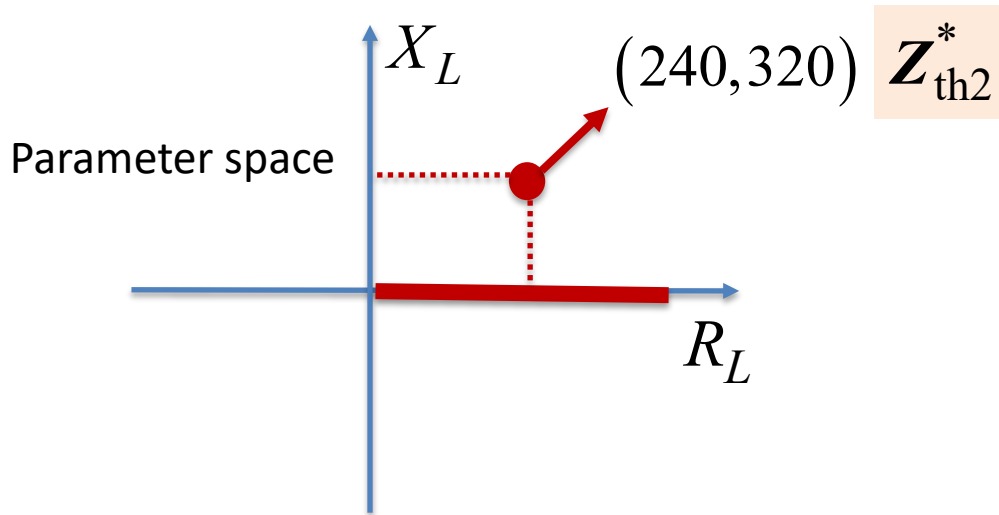
$$V_{th} = 60\sqrt{2} \angle 45^\circ$$

$$Z_{th2} = (240 - 320j) \Omega$$

Question: The variable resistor in the circuit shown in Fig. P10.46 is adjusted until the average power it absorbs is maximum.

a) Find R .

Find the condition of R_L



Phase-restrictive load

$$\varphi_L = 0$$

Recall the condition

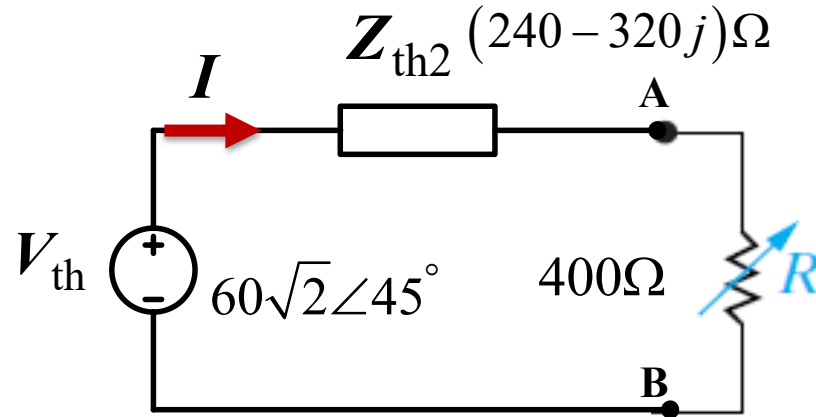
$$|Z_L| = |Z_{Th}|$$

$$Z_{th2} = (240 - 320j)\Omega$$

The resulting Load

$$R_L = |Z_{th2}| = \sqrt{240^2 + (-320)^2} = 400\Omega$$

b) Find the maximum average power.



$$I = \frac{V_{th}}{Z_{th} + R_L} = \frac{60\sqrt{2}\angle 45^\circ}{240 - 320j + 400} = 0.12\angle 71.56^\circ$$

$$P_L = |I|^2 R_L = 0.12^2 \times 400 = 5.62W$$

c) Find a resistor in Appendix H that would have the most average power delivered to it.

Resistors (5% tolerance)[Ω]					
10	100	1.0 k	10k	100k	1.0 M
	120	1.2 k	12 k	120 k	
15	150	1.5 k	15 k	150 k	1.5 M
	180	1.8 k	18 k	180 k	
22	220	2.2 k	22 k	220 k	2.2 M
	270	2.7 k	27 k	270 k	
33	330	3.3 k	33 k	330 k	3.3 M
	390	3.9 k	39 k	390 k	
47	470	4.7 k	47 k	470 k	4.7 M
	560	5.6 k	56 k	560 k	
68	680	6.8 k	68 k	680 k	6.8 M

Theoretical value

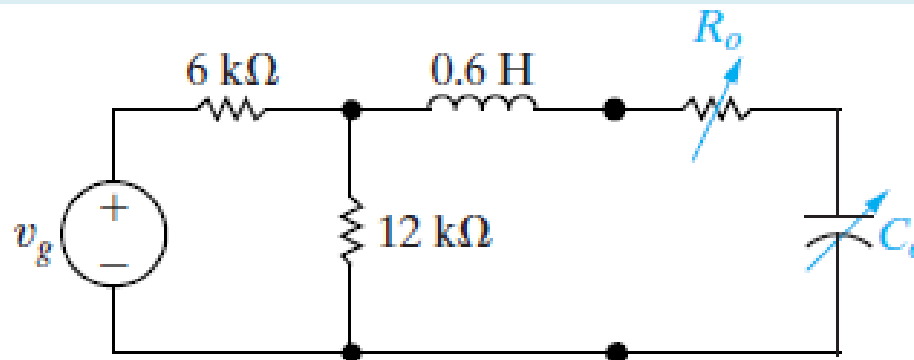
$$R_L = 400\Omega$$

Exercise (P.48)

The peak amplitude of the sinusoidal voltage source in the circuit shown in Fig. P10.48 is **180 V** and its frequency is **5000 rad/s**. The **load resistor** can be varied from **0 to 4000 ohm**, and the **load capacitor** can be varied from **0.1 uF to 0.5uF**.

Questions:

- Calculate the average power delivered to the load when $R_o = 2000 \text{ ohm}$ and $C_o = 0.2 \mu\text{F}$.
- Determine the settings of R_o and C_o that will result in the most average power being transferred to R_o .

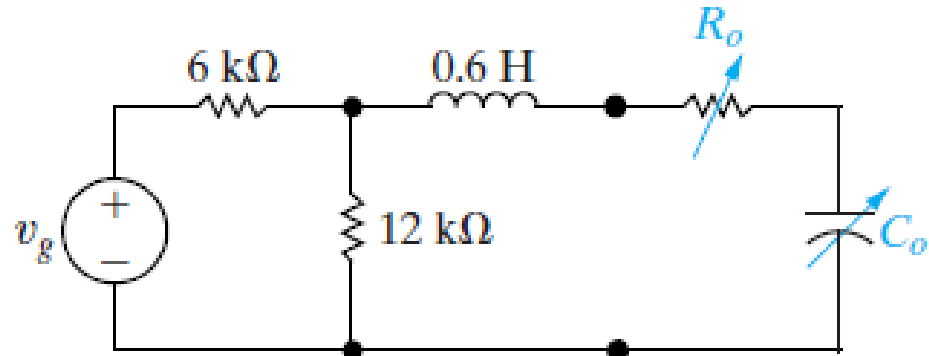


The peak amplitude of the sinusoidal voltage source in the circuit shown in Fig. P10.48 is **180 V** and its frequency is **5000 rad/s**. The **load resistor** can be varied from **0 to 4000 ohm**, and the **load capacitor** can be varied from **0.1 uF to 0.5uF**.

$$v_g = 180 \cos 5000t$$

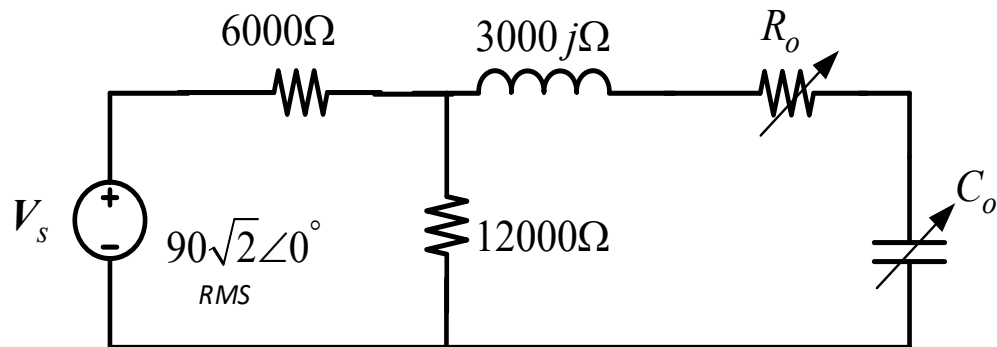
$$R_o \in [0, 4000\Omega]$$

$$C_o \in [0.1\mu F, 0.5\mu F]$$



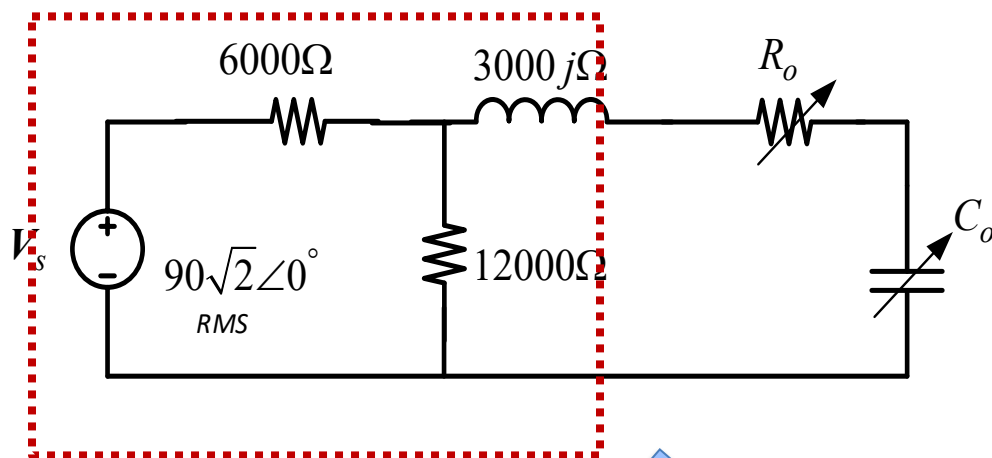
$$R_o \in [0, 4000\Omega]$$

$$X_C \in [-2000j\Omega, -400j\Omega]$$



Questions:

a) Calculate the average power delivered to the load when $R_o = 2000 \text{ ohm}$ and $C_o = 0.2 \mu\text{F}$.

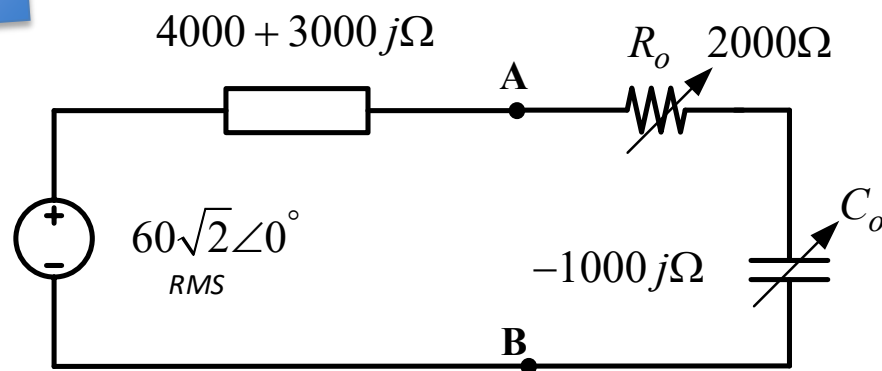


$$X_C \in [-2000j\Omega, -400j\Omega]$$

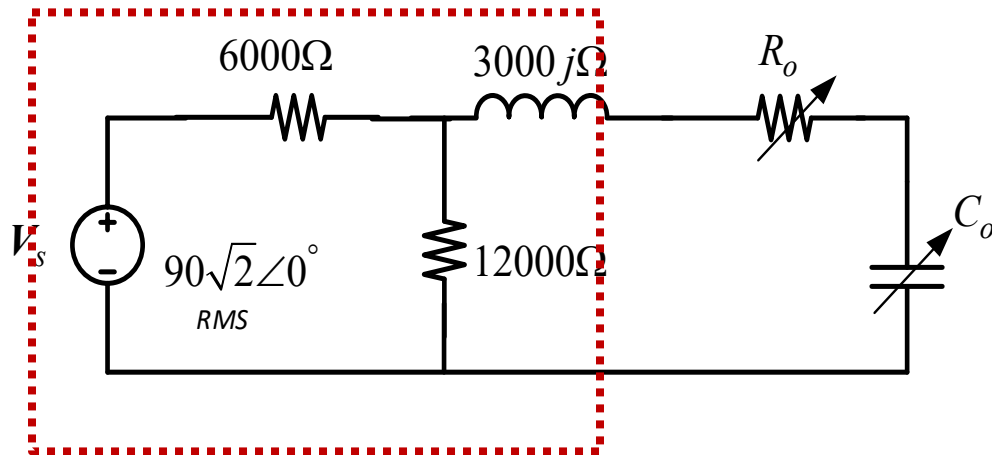
$$R_o \in [0, 4000\Omega]$$

Step 1: Thevenin Equivalent

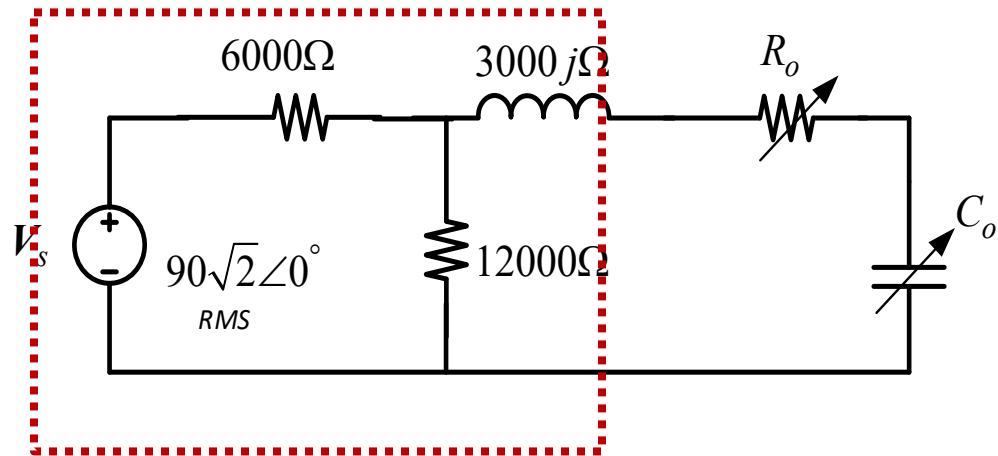
$$X_c = -\frac{1}{\omega C} = -1000$$



Thevenin equivalent V_{op}



Z_{Th}

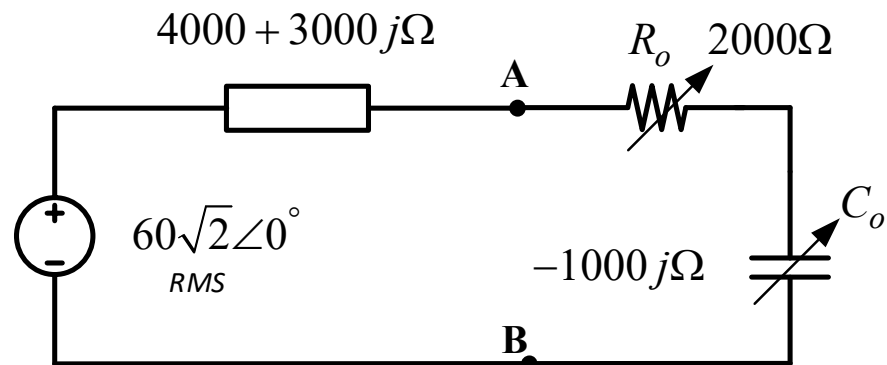


Questions:

a) Calculate the average power delivered to the load when $R_o = 2000 \text{ ohm}$ and $C_o = 0.2 \mu\text{F}$.

Step 2:

$$I = \frac{60\sqrt{2}}{4000 + 3000j + 2000 - 1000j}$$
$$= 13.4 \text{ mA} \angle -18.43^\circ$$



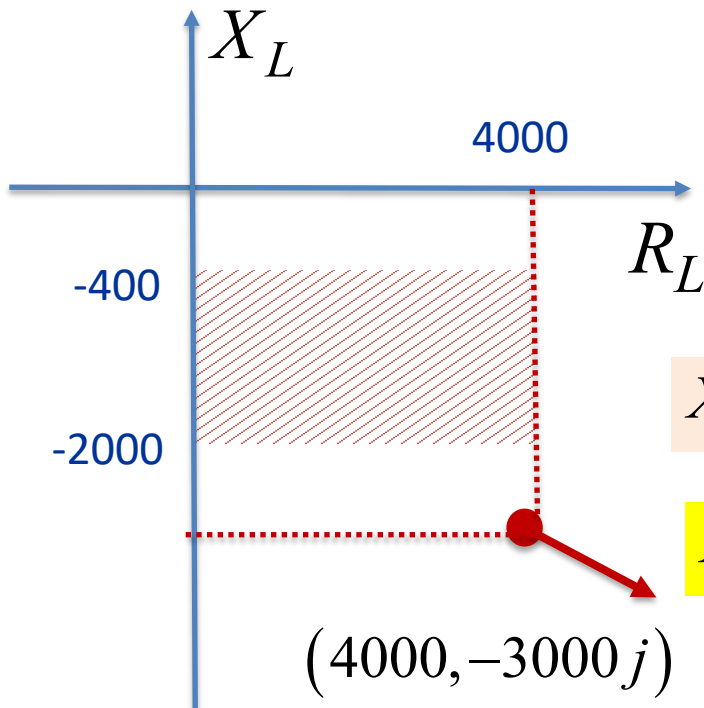
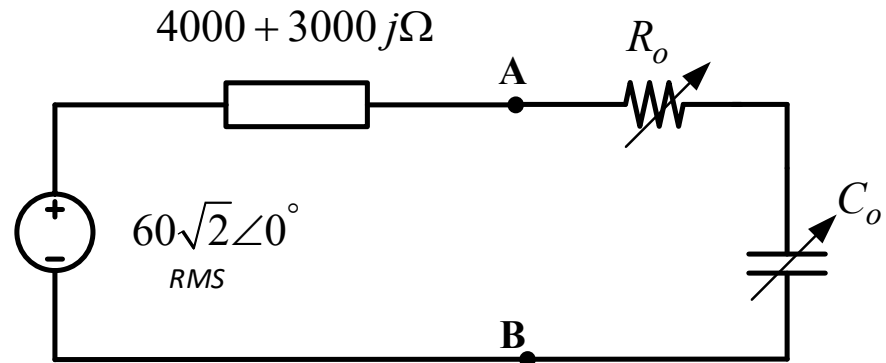
Step 3:

$$P_{R_o} = |I|^2 R_o = (13.4 \times 10^{-3})^2 2000 = 0.359 \text{ W}$$

Questions:

b) Determine the settings of R_o and C_o that will result in the most average power being transferred to R_o .

$$\begin{cases} R_o \in [0, 4000\Omega] \\ X_C \in [-2000j\Omega, -400j\Omega] \end{cases}$$



$$X_L \rightarrow -X_{Th}$$

$$X_C = -2000\Omega$$

$$R_L = \sqrt{R_{Th}^2 + (X_L + X_{Th})^2}.$$

$$\begin{aligned} &= \sqrt{4000^2 + (-2000 + 3000)^2} \\ &= 4123\Omega > 4000\Omega \end{aligned}$$

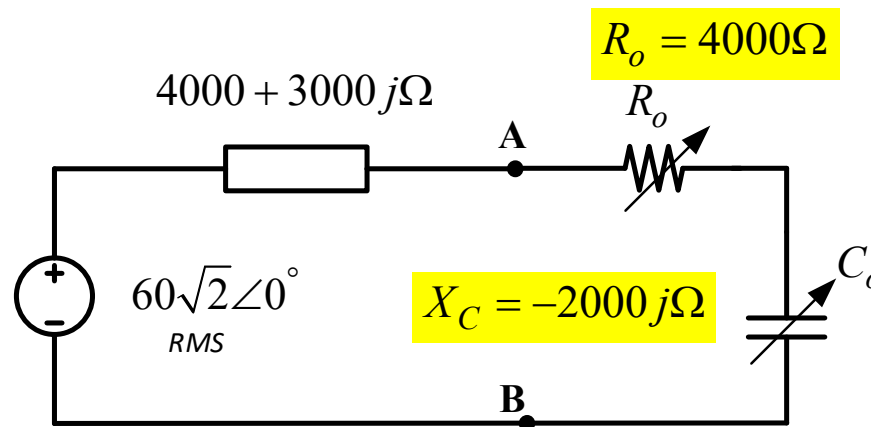
$$R_o = 4000\Omega$$

Questions:

c) What is the average power in (b)? Is it greater than the power in (a)?

Step 1:

$$I = \frac{60\sqrt{2}}{4000 + 3000j + 4000 - 2000j}$$
$$= 10.52\text{mA} \angle -7.125^\circ$$



Step 2:

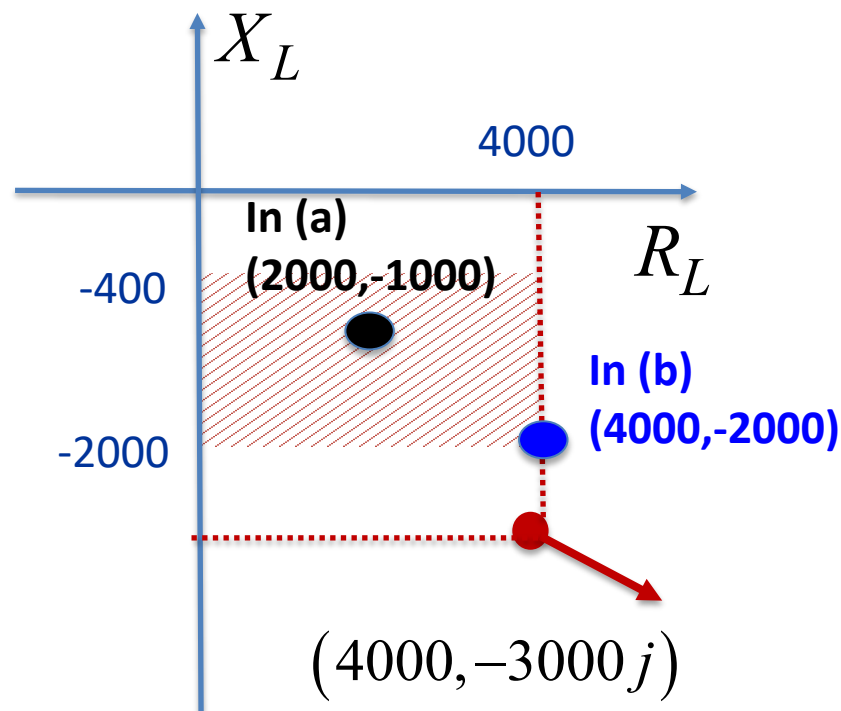
$$P_{R_o} = |I|^2 R_o = (10.52 \times 10^{-3})^2 4000 = 0.443\text{W}$$

Questions:

c) What is the average power in (b)? Is it greater than the power in (a)?

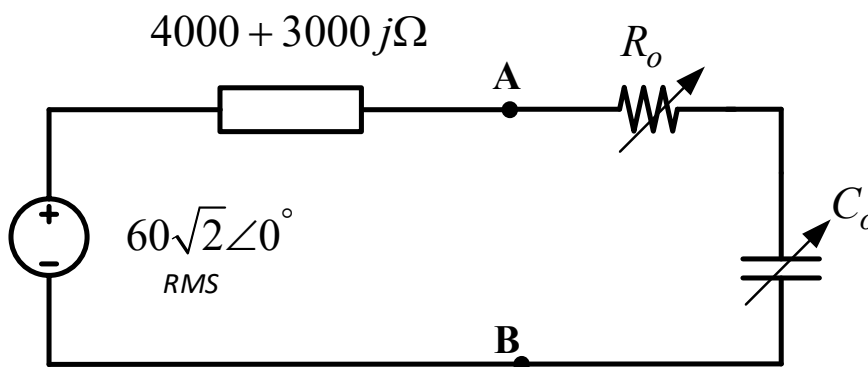
$$P_{Ro} = 0.443W \quad > \quad 0.359W$$

in (b) in (a)



Questions:

- d) If there are no constraints on R_o and C_o what is the maximum average power that can be delivered to a load?
- e) What are the values of R_o and C_o for the condition of (d)?



$$\mathbf{Z}_L = \mathbf{Z}_{th}^* = 4000 - 3000j\Omega \quad P_{\max} = \frac{|V_{th}|^2}{4R_L} = \frac{(60\sqrt{2})^2}{4 \times 4000} = 0.45W$$

$$\left\{ \begin{array}{l} R_o = 4000\Omega \\ X_o = -3000\Omega \end{array} \right. \Rightarrow C_o = -\frac{1}{X\omega} = -\frac{1}{-3000 \times 5000} = 0.067\mu F$$

Questions:

f) Is the average power calculated in (d) larger than that calculated in (c)?

In (c)

$$P_{Ro} = |I|^2 R_o = (10.52 \times 10^{-3})^2 4000 = 0.443W \quad \text{restrictive load}$$

In (d)

$$P_{\max} = \frac{|V_{th}|^2}{4R_L} = \frac{(60\sqrt{2})^2}{4 \times 4000} = 0.45W \quad \text{non-restrictive load}$$

Yes, larger than that in (c)

End